Chapter 7.

ABOUT THIS TEMPLATE

Contents

7.1 Document Class and Options 35
7.2 Commands for Setup 36
7.3 Commands for Display 40
7.4 Emphasis of Author Names in Bibliography 41
7.5 Double-blind Review Format 42
7.6 Preloaded Class and Packages 43
7.7 History 43

7.1 Document Class and Options

This is the MPU Doctoral Thesis Template, Version *7.8.1*, released on Date *2026-05-06*. To use this template, we start a LaTeX document by `\documentclass{mpThesis}`. This document class defaults with options: `Libertine`, `PCOM`, `Proposal`, `Review`, `Screen`, `NoHardback`, `OneHalfSpacing`, and `PhDAfterName`. The available options are categorized as follows.

Font families

- `CM` — The classic Computer Modern family.
- `Times` — The Times Roman family.
- `Utopia` — The Utopia family.
- `Garamond` — The EBGaramond family.
- `Palatino` — The Palatino family.
- † `Libertine` — The Libertine family.

Programmes

- † `PCOM` — Computer Applied Technology.
- `PAIDD` — Artificial Intelligence driven Drug Discovery.
- `PETI` — Educational Technology and Innovation.
- `JUC` — Computer Applied Technology Joint Programme with University of Coimbra.
- `JUB` — Computer Applied Technology Joint Programme with University of Bologna.

Thesis types

† `Proposal` — Research proposals for candidature confirmation.
• `Thesis` — Theses for thesis defences.

Thesis display formats

† `Blind` — Hiding the `BlindRegion` environment and `\BlindText`, and disabling the emphasis of author names, to support the double-blind peer review.
† `Review` — Review format with colored links and vertical rulers.
• `Final` — Final plain format with QR code and defence panel information, no color links and no line numbers.
† `Screen` — Equal inner and outer margins, for displaying on screen.
• `Print` — Larger inner margins. for printing and binding.
† `NoHardback` — No spine text and no calligraphic Chinese MPU watermark.
• `Hardback` — Formal binding format, with spine text and calligraphic Chinese MPU watermark.
† `OneHalfSpacing` — 1.5× line spacing in text.
• `DoubleSpacing` — 2× line spacing in text.
† `PhDAfterName` — "Ph. D." below the student name on the hardcover.
• `NoPhDAfterName` — No "Ph. D." below the student name on the hardcover.

Algorithm packages

- `AlgoMic` — Load the traditional `{algorithm}` and `{algorithmic}` packages.
- `AlgoX` — Load the improved `{algorithm}`, `{algorithmicx}` and `{algpseudocode}` packages.
- `AlgoIIe` — Load the improved and combined `{algorithm2e}` package.
- `MathPar` — Load the `{mathpartir}` package for type setting inference rules.

The `AlgoMic`, `AlgoX` and `AlgoIIe` options are mutually exclusive.

7.2 Commands for Setup

Student and supervisor

- `\StudentName[<break name>]{<full name>}`
  - Purpose: Sets the student's full name and an optional break-friendly version.
  - Arguments:
    * `#1` (optional): Name with line breaks.
    * `#2` (required): Full name of the student.

  - Example: `\StudentName[Brianchon\\Pascal]{Brianchon Pascal}`

- `\StudentNumber{<student number>}`
  - Purpose: Stores the student's identification number.
  - Argument: `#1`: Student number.
  - Example: `\StudentNumber{P-12-1234-1}`

- `\Supervisor{<name>}`
  - Purpose: Sets the name of the supervisor.
  - Argument: `#1`: Name of the supervisor.
  - Example: `\Supervisor{Desargues Monge}`

- `\JointSupervisor{<name>}`
  - Purpose: Sets the name of the joint supervisor.
  - Argument: `#1`: Name of the joint supervisor.
  - Example: `\JointSupervisor{Euler Gauss}`

- `\CoSupervisor{<name>}`
  - Purpose: Sets the name of the co-supervisor, if any.
  - Argument: `#1`: Name of the co-supervisor.
  - Example: `\CoSupervisor{Hilbert Minkowski}`

Date and time

- `\Date{<YYYY-MM-DD>}`
  - Purpose: Sets the document date.
  - Argument: `#1`: Date in `YYYY-MM-DD` format.
  - Example: `\Date{2025-09-17}`

- `\DateToday`
  - Purpose: Sets the document date to the current date.

- `\DefenceDateTime{<YYYY-MM-DD>}{<HH:MM>}`
  - Purpose: Sets the date and time of the thesis defence.
  - Arguments:
    - `#1`: Date in `YYYY-MM-DD` format.
    - `#2`: Time in `HH:MM` format.
  - Example: `\DefenceDateTime{2025-10-01}{14:30}`

Thesis

- `\Title[<break title>]{<full title>}`

  - Purpose: Sets the thesis title and optional break-friendly version.
  - Arguments:
    * `#1` (optional): Title with line breaks.
    * `#2` (required): Title of the thesis.
  - Example: `\Title[AI\\in Healthcare]{AI in Healthcare}`

- `\ThesisNumber{<seq>}{<programme>}{<unit>}{<lna>}{<ftw>}`
  - Purpose: Sets the thesis number components.
  - Arguments:
    * `#1`: Sequence number.
    * `#2`: Programme code.
    * `#3`: Academic unit code.
    * `#4`: Last name of the student.
    * `#5`: First (non-article) word of the title.
  - Example: `\ThesisNumber{00123}{PCOM}{FCA}{pascal}{sample}`

- `\ThesisKeywords{<keywords>}`
  - Purpose: Sets thesis keywords.
  - Argument: `#1`: List of keywords.
  - Example: `\ThesisKeywords{Artificial Intelligence, Healthcare, Machine Learning}`

- `\Keywords{\InsertThesisKeywords}`
  - Purpose: Displays the stored thesis keywords with the “Keywords” prefix.

Chinese Title and Abstract

- `\ChsTitle{<full title>}`
  - Purpose: Sets the thesis title in Simplified Chinese.
  - Arguments: `#1`: Title of the thesis in Simplified Chinese.
  - Example: `\ChsTitle{论文题目}`

- `\ChtTitle{<full title>}`
  - Purpose: Sets the thesis title in Traditional Chinese.

- `\ThesisChsKeywords{<CHS keywords>}`
  - Purpose: Sets thesis keywords in Simplified Chinese.
  - Argument: `#1`: List of keywords in Simplified Chinese.
  - Example: `\ThesisChsKeywords{人工智能、医疗保健、机器学习}`

- `\ThesisChtKeywords{<CHT keywords>}`
  - Purpose: Sets thesis keywords in Traditional Chinese.

- Environment `ChsAbstract`
  - Purpose: Encloses thesis abstract in Simplified Chinese, and displays the abstract with the Chinese thesis title.
  - Example: `\begin{ChsAbstract}`简体中文摘要`\end{ChsAbstract}`
- Environment `ChtAbstract`
  - Purpose: Encloses thesis abstract in Traditional Chinese, and displays the abstract with the Chinese thesis title.
- `\ChsKeywords{\InsertThesisChsKeywords}`
  - Purpose: Displays the stored Simplified Chinese thesis keywords with the "关键字" prefix.
- `\ChtKeywords{\InsertThesisChtKeywords}`
  - Purpose: Displays the stored Traditional Chinese thesis keywords with the "關鍵字" prefix.

Chinese text compiled by pdfLaTeX is implemented by the `CJKutf8` package. There are only two Simplified Chinese fonts built in the TeX Live distribution. If you install your own CJK Chinese fonts in UTF8 encoding, you can specify the fonts to use in the title and abstract, by commands: `\ChsTitleFont`, `\ChtTitleFont`, `\ChsHeaderFont`, `\ChtHeaderFont`, `\ChsTextFont`, and `\ChtTextFont`. However, you don't usually need to touch these commands.

Panel Each of these sets the name of a panel member:

- `\PanelChair{<name>}`
  - Purpose: Sets name of the panel chair.
- `\PanelExternal{<name>}` | `\PanelExternalA{<name>}`
  - Purpose: Sets name of the external examiner.
- `\PanelJointExternal{<name>}` | `\PanelExternalB{<name>}`
  - Purpose: Sets name of the external examiner nominated by the joint university.
- `\PanelInternal{<name>}` | `\PanelInternalA{<name>}` | `\PanelExaminer{<name>}` | `\PanelExaminerA{<name>}`
  - Purpose: Sets name of the internal examiner.
- `\PanelJointInternal{<name>}` | `\PanelInternalB{<name>}` | `\PanelExaminerB{<name>}`
  - Purpose: Sets name of the second internal examiner, or the internal examiner nominated by the joint university.

7.3 Commands for Display

These commands retrieve stored values for use in the document:

Student and supervisor

- `\InsertStudentName`
- `\InsertStudentNameBreak`
- `\InsertStudentNumber`
- `\InsertSupervisor`
- `\InsertJointSupervisor`
- `\InsertCoSupervisor`

University

- `\InsertUniversity`
- `\InsertJointUniversity`
- `\InsertAcademicUnit`
- `\InsertAcademicShort`

Programme

- `\InsertProgramme`
- `\InsertProgrammeBreak`
- `\InsertJointProgramme`
- `\InsertJointProgrammeBreak`
- `\InsertProgrammeCode`

Date and time

- `\InsertDate`
- `\InsertLongDate`
- `\InsertMonthYear`
- `\InsertYear`
- `\InsertDefenceDate`
- `\InsertDefenceTime`

Thesis

- `\InsertTitle`
- `\InsertTitleBreak`
- `\InsertThesisNumber`

- \InsertThesisNumberBreak
- \InsertThesisNumberStyle
- \InsertThesisNumberBreakStyle
- \InsertThesisUrl
- \InsertThesisKeywords
- \InsertThesisChsKeywords

Panel

- \InsertPanelChair
- \InsertPanelExternal
- \InsertPanelJointExternal
- \InsertPanelExternalA
- \InsertPanelExternalB
- \InsertPanelInternal
- \InsertPanelJointInternal
- \InsertPanelInternalA
- \InsertPanelInternalB
- \InsertPanelExaminer
- \InsertPanelExaminerA
- \InsertPanelExaminerB

About this thesis class

- \InsertThesisClassVersion — “7.8.1”
- \InsertThesisClassDate — “2026-05-06”
- \InsertThesisClassDisplayCode — “MPU Thesis Template v7.8.1”

7.4 Emphasis of Author Names in Bibliography

The customized {mpUnsrt.bst} bib-style provides code to scan the author list of a bib-entry to find and highlight the specified author names. The .bst style file is programmed by a postfix language on a stack machine, which is quite interesting to look at.

To specify the author names to emphasize, we need to add entries in the .bib file with category @emphasized. Only two fields are allowed in an entry of this category: author and note. The author field specifies the author list as in a normal category, while the note field specifies the highlighting style, which must be a command taking one argument of the author name, \textbf by default. For example:

```
@emphasized{emphasized/author/list/3,
  author = {Wei Ke and David Hilbert and Gaspard Monge},
```

```
  note    = {\textbf}
}
```

There can be many entries of this category, and one entry must be selected to take effect. To select the entry of the author list, we write `\EmphasizeAuthors{<entry-key>}`. For the above example:

```
\EmphasizeAuthors{emphasized/author/list/3}
```

This must be done before any bib-entries and the inclusion of the `.bst` and `.bib` files.

By the way, we can use command `\bibentry{<entry-key}` to display a bib-entry anywhere in the thesis outside the bibliography list. This is particularly useful for listing the publication produced for the thesis defence, usually in the introduction. See Section 1.3.1 for example.

7.5 Double-blind Review Format

To support the double-blind peer review, option `Blind` hides the student and supervisor names. The option also disables the double degree modes. There are still other sensitive texts that can be used to identify the student and supervisor. They must also be hidden. Such text can be enclosed within environment `BlindRegion`:

```
\chapter{Acknowledgements}\label{ch:ack}

\begin{BlindRegion}
... paragraphs ...
\end{BlindRegion}

\subsection{Current Publication}\label{sec:publication}

\begin{BlindRegion}
\begin{enumerate}[label=(C\arabic*)]
  \item \bibentry{ke2009graph}
\end{enumerate}
\end{BlindRegion}

\chapter{Biography}\label{app:biography}

\begin{BlindRegion}
... paragraphs ...
\end{BlindRegion}
```

Or, such short text can be put in the `\BlindText` macro:

```
#    (c) 2021 \BlindText{Wei Ke and Ka-Hou Chan}
# Id: pp.py 4838 2021-11-14 12:02:46Z \BlindText{wke@IPM.EDU.MO}
```

Then, if option `Blind` is selected, the enclosed text is hidden, otherwise shown as normal. At least

1. the *Acknowledgements* chapter,
2. the publication list, and
3. the *Biography* appendix

must be enclosed in the blind regions. There are also other places that the author information can occur, for example, in the copyright declaration of your implementation code.

While the `Blind` option disables the emphasis of author names in the bibliography, including the `\bibentry` items, the BibTeX may not be able to recompile automatically, you need to remove all the auxiliary files to let the recompilation happen.

7.6 Preloaded Class and Packages

These LaTeX class and packages are loaded by the template:
`[12pt,a4paper]{book}`,
`{geometry}`, `[T1]{fontenc}`, `{CJKutf8}`, `{amsmath,amsthm,amssymb}`, `{mathtools}`,
`{makecell}`, `{array}`, `{bm}`,
`[dvipsnames,x11names,svgnames]{xcolor}`, `[pdftex]{hyperref}`,
`{multirow,multicol}`, `{wrapfig}`, `{graphicx}`, `{caption,subcaption}`,
`[style=long4col,toc,automake]{glossaries}`, `[noindentafter]{titlesec}`, `{titletoc}`,
`{tocbibind}`, `{notoccite}`, `{setspace}`, `{fancyhdr}`, `{etoolbox}`, `{stmaryrd}`, `{marvosym}`,
`[en-GB,calc]{datetime2}`, `{varwidth}`, `{keyval}`, `{qrcode}`, and `{bibentry}`.

A few packages are optionally loaded:
`{eso-pic}` (option `Hardback`), `{fgruler}` (option `Review`), and `{comment}` (option `Blind`).

7.7 History

v7.8.1 • 6th May 2026

- Fix a typo;

v7.8 • 20th April 2026

- Add Traditional Chinese title/abstract/keyword support:
  `\ChtTitle`, `\ThesisChtKeywords`, `\ChtKeywords`, and environment `ChtAbstract`;

v7.7 • 27th March 2026

- Replace line numbers with side rulers;
- Add the `Blind` option, together with `BlindRegion` and `\BlindText` to support double-blind peer review;
- Add the `\CoSupervisor` and `\InsertCoSupervisor` commands to set and display the official co-supervisor of a student, if any;
- Add macros about this thesis class — version number, release date, and display code;

v7.6 • 6th March 2026

- Add Simplified Chinese title/abstract/keyword support:
  `\ChsTitle`, `\ThesisChsKeywords`, `\ChsKeywords`, and environment `ChsAbstract`;
- Add a thesis skeleton, with brief descriptions of what are expected in the individual chapters;

v7.5 • 10th October 2025

- fix the missing package bug with the `Hardback` option;
- Pack all logos into one PDF file and use page selection in `\includegraphics` to load individual logos;
- Add a research proposal skeleton, with brief descriptions of what are expected in the individual chapters;

v7.4 • 30th September 2025

- options to include and integrate various algorithm packages: `algorithm`, `algorithmic`, `algorithmicx`, `algpseudocode`, `algorithm2e` and `mathpartir`;
- new `sample-algo.tex` to test the included algorithm packages;

v7.3 • 18th September 2025

- new (`Review` mode) class option `Screen` (opposite: `Print`) to turn off odd/even page margin difference, better for screen display;
- new sample thesis modified from a real published paper to show more LaTeX type setting examples;
- restart line numbers from 1 for every page;

- add a chapter to list the document class options and the commands provided;
- move the version history to the above chapter;
- fix some `lineno` issues in display math, and extend line numbers to appendices;

v7.2 • 13th August 2025

- new set of panel member definitions for joint-degree programmes,
  `\PanelChair`
  `\PanelExternal` or `\PanelExternalA`
  `\PanelJointExternal` or `\PanelExternalB`
  `\PanelInternal` or `\PanelInternalA` or `\PanelExaminer` or `\PanelExaminerA`
  `\PanelJointInternal` or `\PanelInternalB` or `\PanelExaminerB`;
- fine tune the logos for joint programme;
- adjust signature page;
- remove the MPI glossary;

v7.1 • 19th June 2025

- add `\PanelExaminerC` and `\PanelExaminerD` for joint programme;
- adjust signature page;

v7.0 • 16th June 2025

- add `\InsertUniversity` and `\InsertJointUniversity`;
- add `\JointSupervisor` for use with joint programme;
- adjust the font size and rewrite the statement from the cover page;
- refine “mpThesis.cls”;

v6.4 • 23rd May 2025

- provide descriptions and examples for various options of “mpThesis”;
- address the code for entering long name of author/title/programme;
- Co-supervisor will not be showed;
- add `\InsertAcademicShort` for the use of ThesisNumber;
- add `\InsertProgrammeCode` for the use of ThesisNumber;

v6.3 • 21st May 2025

- add options of `PCOM`, `PAIDD`, `PETI` for Ph.D. programmes;
- add option of `JUC`, `JUB` for joint programmes;
- resize/adjust the MPU logo(s) in FrontMatter;
- providing the UC logo for the use of FCA-JUC programme;

- rewrite the definition of “font families” in .cls;
- rewrite the definition of “thesis types” in .cls;
- rewrite the definition of “produce modes” in .cls;
- rewrite the of **ACKNOWLEDGMENTS**;

v6.2 • 25th March 2024

- enable “x11names” from package “xcolor”;
- redraw MPU-FCA-logo.pdf;

v6.1 • 3rd April 2025

- use CalliCover as the default `Hardback` paper;

v6.0 • 24th March 2023

- add `Hardback` option for final printing;
- add “copyright page” in between “cover page” and “title page”;
- redesign “cover page”, “title page”, and “declaration page”;
- “declaration page” is created as a *.tex file and is allowed to be modified;
- `\DefenceDate`, `\PanelChair`, `\PanelExternal` and `\PanelExaminer` information needs to be entered;
- supports optional long title names (with line break) to be displayed on the copyright page only;
- a template of “Author Biography” is provided;
- make font “Libertine” and cover “CalliCover” the default;
- move “Ethics Checklist” to appendix;
- latex (macros) code refinement;

v5.9 • 6th January 2023

- support “subsubsection” indexing;
- “notoccite” package is enabled by default in “mpThesis.cls”;

v5.8 • 4th January 2023

- a blank page has been added at the end of thesis;
- fixed issue from package “glossaries”;
- fixed template mark displaying in “Final” mode, now only on the back of the cover page;
- added the `DoubleSpacing` option to enable double spacing when needed, the default is one-half spacing;
- added the `NoTocCite` option to include the “notoccite” package to prevent the numbering of bibliography from being affected by the citations in TOC;

v5.7 • 27th October 2022

- restrict package “glossaries” using version 4.46;
- define the (MPU and MPI glossary by default to eliminate the quirk in the “glossaries” package;
- determine double-spacing in mainMatter;
- increase TOC depth level to 2;
- show the “url” and “doi” items in the “mpUnsrt” style.
- a sample of “publication list” is provided in Chapter 1;
- section refinement in Chapter 1 to Chapter 3;
- adjust “citecolor”;

v5.6 • 22nd September 2022

- made “List of Figures/Tables/Algorithms/Abbreviations” optional;
- adjusted the position of the example section;
- provided more advanced examples;
- reformatted section “Template Versioning”;
- release to canvas;

v5.5 • 20th September 2022

- added “List of Algorithms”;
- provided an example of algorithm in Chapter 4;
- added optional long name (with line breaks) support, including student names and programme names, shown only on the title page;
- `NoToC` option;
- adjusted the vertical position of the Chinese calligraphic MPU logo in page headers;
- adjusted vertical spacing of the table on the title page;

v5.4 • 15th September 2022

- refined title page;
- adjusted spacing;
- added Chinese MPU logo (in .pdf);
- redesigned header format;

v5.3 • 13th September 2022

- refined format of “mpUnsrt.bst”;
- added emphasized authors support to “mpUnsrt.bst”;
- improved TOC macros;

v5.2 • 6th September 2022

- added reference style “mpUnsrt.bst”;
- added more font families;
- disposed “IndentFirst”;
- improved template manuals;
- make minitoc singlespacing;
- provided two types of `Thesis` or `Proposal`, `Thesis` is default;
- provided two modes of `Review` or `Final`, `Review` is default;
- `Review` mode shows hyperlinks and line numbers;
- `Final` mode removes line numbers, no color for hyperlinks;
- minor font size adjustment;
- added template version marks to even pages;

v5.1 • 2nd September 2022

- Overleaf online tested;
- online version refinement;
- made “IndentFirst” optional;
- redefined the cross-reference prefix;
- aligned list of abbreviations;

v5.0 • 29th August 2022

- major revision;
- item definition of template classes and refinement of LaTeX code;
- regeneration of MPU logo;
- provide more font family;
- an example about profile updating is provided;

v4.8 • 24th August 2022

- add mini-tables-of-contents in each chapter;
- move “Acknowledgments” to frontmatter;
- add “List of Abbreviations”;
- provide an example of adding glossary in Chapter 3;
- add the section of “Template Versioning” in Chapter 6;
- the first indent have been changed to default;
- latex (macros) code refinement;

v4.2 • 8th March 2022

- add “Universidade Politécnica de Macau” on the Cover page;
- cover page redesign;
- regenerate the optimized MPU loge;
- refine TOC format;

v4.1 • 4th March 2022

- appendix header correction;
- section header alignment;

v4.0 • 7th March 2022

- update “MPI” to “MPU”;
- update “School” to “Faculty”;
- adjust the size and position of the MPU logo.
- formatting student ID: `P-XX-XXXX-X`;
- change to use small cap format for the section header;
- highlight fillable item;
- few formatting adjustments;

v3.7 • 26th November 2021

- dispose the front of “Times New Roman”;
- update title page;
- provide the logo in PDF format;
- few formatting adjustments;

v3.5 • 13th April 2021

- update TOC’s chapter and title;
- update the chapter of “Acknowledgements”;
- provide an example of using equation in Chapter 1;
- provide an example of using table in Chapter 2;
- provide an example of adding figure in Chapter 3;
- provide an example of using list in Chapter 5;
- provide an example of citing literature in Chapter 5;
- other minor adjust to format;

v3.1 • 27th January 2021

- update cover paper statement;
- change citation style to ieeetr style;
- remove chinese abstract;
- few formatting adjustments;

v3.0 • 15th January 2021

- format TOC;
- change first indent to 1.2cm;
- change header/footer font size to 10;
- other adjustments;

v2.3 • 7th January 2021

- supported chinese abstract;
- adjust TOC;

v2.1 • 6th January 2021

- update cover pager;
- design and formatting;

v1.0 • 4th January 2021

- first release;